Example 1

- Pinkie Pie is curious if party goers prefer chocolate cake over vanilla cake. At her latest party, she surveys 75 ponies (**party_data**). Each pony is asked which flavor they prefer (*preference*).
- She ultimately wants to determine if more than 50% prefer chocolate cake.
- Let's now formally test her hypothesis at the $\alpha = 0.05$ level.

```
party_data |> one_proportion(outcome = preference,
                             event = "Chocolate",
                             p = 0.5,
                             alternative = "greater",
                             alpha = 0.05)
```

```
One-sample proportion

p̂ = 0.7067 (53/75)

95% CI for π: (0.6036, 0.8097)

Hypotheses:
H₀: π ≤ 0.5
H₁: π > 0.5
Test statistic: z = 3.58
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Example 2

- Fluttershy has a hunch: younger foals might be less likely to share their candy on Nightmare Night compared to older foals. To test this, she observes foals from each age group (*age_group*) and records whether each one shared their candy (*shared_candy*). The data is recorded in **candy_data**.
- Let's test Fluttershy's hunch at the $\alpha = 0.10$ level. How should we update this code?

```
candy_data |> independent_proportions(outcome = shared_candy,
                                       grouping = age_group,
                                       event = "Yes",
                                       p = 0,
                                       alternative = "greater",
                                       alpha = 0.1)
```

```
Independent proportions

p̂[Older] = 0.8167
p̂[Young] = 0.4167
Point estimate: p̂[Older] - p̂[Young] = 0.4

90% CI for π[Older] - π[Young]: (0.2669, 0.5331)

Hypotheses:
H₀: π[Older] - π[Young] ≤ 0
H₁: π[Older] - π[Young] > 0
Test statistic: z = 4.51
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.100)
```

Let's Practice! 1

Rainbow Dash wants at least 60% of Cloudsdale weather reports to arrive on time. She checks a random sample of 120 reports and records whether each arrived on time in `on_time`. Is there evidence that more than 60% of reports arrive on time?

```
weather_reports |> one_proportion(outcome = on_time,
                                   event = "Yes",
                                   p = 0.60,
                                   alternative = "greater")
```

```
One-sample proportion

p̂ = 0.6333 (76/120)

95% CI for π: (0.5471, 0.7196)

Hypotheses:
H₀: π ≤ 0.6
H₁: π > 0.6
Test statistic: z = 0.75
p-value: 0.228
Conclusion: Fail to reject the null hypothesis (p = 0.228 ≥ α = 0.050)
```

Let's Practice! 2

Applejack believes that more than half of the apples from a new orchard section meet her quality standard. She randomly selects 100 apples and records `meets_standard` as `Yes` or `No`. Is there evidence that more than 50% of apples from this section meet the standard?

```
apple_quality |> one_proportion(outcome = meets_standard,
                                event = "Yes",
                                alternative = "greater")
```

```
One-sample proportion

p̂ = 0.67 (67/100)

95% CI for π: (0.5778, 0.7622)

Hypotheses:
H₀: π ≤ 0.5
H₁: π > 0.5
Test statistic: z = 3.4
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

Let's Practice! 3

Twilight Sparkle tries two ways of sharing an exam study guide. One group of 80 students receives a printed guide; a separate group of 80 receives a magical guide. After studying, each student reports whether they feel prepared for the exam (`prepared`). Is the population proportion who feel prepared higher with the magical guide than with the printed guide?

```
study_guides |> independent_proportions(outcome = prepared,
                                        grouping = guide,
                                        event = "Yes",
                                        alternative = "greater")
```

```
Independent proportions

p̂[Magical] = 0.725
p̂[Printed] = 0.525
Point estimate: p̂[Magical] − p̂[Printed] = 0.2

95% CI for π[Magical] − π[Printed]: (0.0532, 0.3468)

Hypotheses:
H₀: π[Magical] − π[Printed] ≤ 0
H₁: π[Magical] − π[Printed] > 0
Test statistic: z = 2.61
p-value: 0.004
Conclusion: Reject the null hypothesis (p = 0.004 < α = 0.050)
```

Let's Practice! 4

Fluttershy observes two separate groups of 70 visitors to her animal sanctuary. One group visits in the morning and the other in the afternoon. For each visitor, she records whether they saw a rabbit (`saw_rabbit`). Is there evidence that the population proportions who see a rabbit differ between morning and afternoon visits?

```
garden_visits |> independent_proportions(outcome = saw_rabbit,
                                          grouping = visit_time,
                                          event = "Yes")
```

```
Independent proportions

p̂[Afternoon] = 0.5714
p̂[Morning] = 0.6286
Point estimate: p̂[Afternoon] − p̂[Morning] = -0.0571

95% CI for π[Afternoon] − π[Morning]: (-0.2192, 0.1049)

Hypotheses:
H₀: π[Afternoon] − π[Morning] = 0
H₁: π[Afternoon] − π[Morning] ≠ 0
Test statistic: z = -0.69
p-value: 0.490
Conclusion: Fail to reject the null hypothesis (p = 0.490 ≥ α = 0.050)
```